

Chalaka Samith

COMPUTER ENGINEERING UNDERGRADUATE

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About Me

Computer Engineering undergraduate with a First Class (GPA 3.971/4.00) and hands-on experience in full-stack development, cloud-native infrastructure, DevOps Concepts, SRE, and Agentic AI. Currently interning as an SRE at WSO2, building AI-driven tooling to automate operational and compliance workflows. Passionate about building systems end-to-end from code to cloud and automating complex workflows with AI agents.

Education

University of Peradeniya <i>BSc.Eng(Hons.) in Computer Engineering, Final GPA: 3.971/4.00 - First Class</i>	Peradeniya, Sri Lanka 2022 – Present
Royal College <i>G.C.E. Advanced Level - Physical Science, 3As Z-score: 2.0291</i>	Colombo 07, Sri Lanka 2018 – 2020

Technical Skills

- **Languages:** Python, Java, JavaScript, TypeScript, C/C++, SQL
- **Frontend & Mobile:** React, React Native, HTML, CSS
- **Backend Frameworks:** Spring Boot, FastAPI, Node.js
- **Cloud & DevOps:** AWS, Azure, Docker, Kubernetes, Terraform, Helm, Jenkins, GitHub Actions, NGINX, Linux
- **Observability & Data:** Prometheus, Grafana, Loki, MySQL, PostgreSQL, MongoDB, Redis
- **AI / ML & Agentic AI:** LangChain, RAG Pipelines, TensorFlow, NumPy, OpenCV, Browser-Use
- **Messaging & APIs:** Apache Kafka, REST APIs, MQTT, AWS IoT Core
- **Tools:** Git, GitHub, Claude Code, Postman, IntelliJ, VS Code, Antigravity, Apache NetBeans

Projects

- Compliance Evidence Submission Portal (WSO2 Intern Project)**  Repo
 - Built a **standalone, full-stack Compliance Evidence Collection Application** at WSO2 that integrates with the organization's open-source GRC platform via its exposed REST API, enabling automated evidence gathering and submission without modifying the core GRC system.
 - Engineered an **Agentic AI** module using **Python** and **Browser-Use** (Chromium browser automation) with **Azure OpenAI LLMs**, where AI agents autonomously navigate cloud consoles and GitHub repositories to collect and structure compliance evidence end-to-end.
 - Developed the **FastAPI** backend (Python) and **TypeScript** frontend; deployed on **WSO2 Choreo** with **Asgardeo** providing enterprise **OAuth 2.0 / OIDC** authentication across organizations.
 - **Technologies:** Python, FastAPI, TypeScript, React, Azure OpenAI LLMs, Browser-Use, Chromium Automation, WSO2 Choreo, Asgardeo, OAuth 2.0 / OIDC, Apache-2.0
- Ops Copilot - SRE Knowledge Base Assistant (WSO2 Intern Project)**
 - Built an internal **RAG (Retrieval-Augmented Generation) pipeline** over WSO2's operational knowledge bases, enabling SRE engineers to query procedures, runbooks, and architecture docs in natural language and receive **source-cited, context-aware answers**.
 - Designed the ingestion pipeline, vector store integration, and LLM prompt engineering to maximize retrieval precision and reduce hallucinations on domain-specific operational content.
 - **Technologies:** RAG, LLM, Azure OpenAI, Vector Database, Python, TypeScript
- CloudCare HMS – AWS Well-Architected & Terraform (Personal Project)**  Repo
 - Designed and deployed a **production-style, AWS-native Hospital Management System** demonstrating the **AWS Well-Architected Framework** across 9 isolated **Terraform stacks** (VPC, ALB, EC2, RDS, Lambda, CloudFront, Route 53, SES, Secrets Manager).
 - Automated deployments via **GitHub Actions** with secure, keyless **OpenID Connect (OIDC)** integration with AWS, eliminating long-lived credentials.
 - **Technologies:** AWS, Terraform, GitHub Actions, OIDC, Docker, FastAPI, React (Vite), IaC
- CloudCare HMS – EKS Microservices & Observability (Personal Project)**  Repo
 - Migrated CloudCare HMS from a monolithic EC2 deployment to **4 containerized FastAPI microservices** orchestrated on **AWS EKS** across development and production namespaces, with **Horizontal Pod Autoscaling (HPA)**.
 - Deployed a full **observability stack** (Prometheus, Grafana, Loki) and managed Kubernetes secrets securely via **AWS Secrets Manager** integration.
 - **Technologies:** AWS EKS, Kubernetes, Terraform, Helm, Prometheus, Grafana, Loki, FastAPI, GitHub Actions

5. LMS University – Full-Stack Digital Education Platform (Personal Project)

Repo

 - Developed a **comprehensive University LMS** bridging Students, Alumni, and Admins, featuring a social feed, real-time private messaging (WebSockets), admin analytics dashboard, and a career portal for alumni job postings and student applications.
 - Built secure **REST APIs** with **Spring Boot**, JWT-based **role-based authentication**, and engineered both a **React web client** and a **React Native/Expo mobile app** with encrypted token storage.
 - Technologies:** Spring Boot, React, React Native (Expo), MySQL, Docker, JWT, WebSockets
6. Explainable AI–Driven Zero-Trust Anomaly Detection for Encrypted Traffic (Final Year Research Project)

 - Designed an **explainable, zero-trust anomaly detection** framework for **encrypted network traffic**, combining **semi-supervised ML** with **Explainable AI (XAI)** methods to identify and interpret security threats while preserving data privacy.
 - Conducted experiments on the **CIC-IDS-2017** dataset; evaluated advanced anomaly detection algorithms and produced interpretability outputs using XAI techniques to support trust and auditability in zero-trust architectures.
 - Technologies:** Python, Semi-Supervised ML, CIC-IDS-2017 Dataset, XAI, NumPy, Matplotlib
7. Lightify – Smart Home IoT Lighting System (Third Year Group Project)

Repo | Web Page

 - Designed and implemented an **intelligent smart home lighting system** detecting **room occupancy** and dynamically adjusting lighting, supported by a **Spring Boot backend**, **MongoDB** data management, and **AWS Cloud** deployment.
 - Programmed **ESP32 controllers** for reliable device communication using **AWS IoT Core (MQTT)**, Bluetooth, and Wi-Fi between embedded devices and the mobile application. Presented at **EngEx 2025**.
 - Technologies:** React Native, Spring Boot, AWS IoT Core, MongoDB, ESP32, MQTT
8. Artery Resolver – Vascular Analysis & Patient Monitoring Software (Second Year Group Project)

Repo

 - Developed a **medical desktop application** for secure, doctor-specific patient management and diagnostic reporting, integrating **Java Swing** UI and **MySQL** database storage.
 - Implemented a **video processing model** using **Python** and **OpenCV** to extract **blood vessel diameter** from medical video data for vascular analysis.
 - Technologies:** Java, Java Swing, Python, OpenCV, MySQL, Video Processing

Experience

WSO2 <i>Site Reliability Engineering Intern, On-site</i>	Colombo, Sri Lanka <i>Apr 2026 – Oct 2026</i>
Working on the SRE team; built and shipped two AI-driven tooling projects (Compliance Evidence Portal & Ops Copilot).	
University of Peradeniya <i>Undergraduate Teaching Assistant, Part-time</i>	Peradeniya, Sri Lanka <i>Sep 2023 – Mar 2026</i>
CO322 Data Structures & Algorithms CO225 Software Construction CO2030 Data Structures & Algorithms II	
University of Peradeniya <i>EngEx – 75th Anniversary Exhibition</i>	Peradeniya, Sri Lanka <i>2025</i>
Presented Lightify (IoT smart lighting) at the University of Peradeniya Engineering Exhibition.	

Professional Development

Terraform - Top 10% Globally (TestDome, May 2026)	AWS Cloud Practitioner Essentials (AWS, 2026)
Introduction to Kubernetes LFS158 (Linux Foundation, May 2026)	MCP: Hands-On with Agentic AI (LinkedIn, Jun 2026)
Mastering RAG: Smart, Data-Driven Apps (IBM, 2026)	Claude Code in Action (Anthropic, Jun 2026)
GitHub Actions for CI/CD (LinkedIn, Jun 2026)	Azure DevOps for Beginners (LinkedIn, Jun 2026)
Getting Started with Jenkins (Simplilearn, 2026)	Learning Ansible (LinkedIn, Jun 2026)
REST API (HackerRank, 2026)	Azure Fundamentals (Simplilearn, 2026)
Software Architecture Foundations (LinkedIn, Jun 2026)	SQL (HackerRank, Jul 2026)

Extra-Curricular Activities

Student Member , Institution of Engineers Sri Lanka (IESL)	2025 – Present
Member , ACES – Association of Computer Engineering Students, University of Peradeniya	2023 – Present
Member , AIESEC – University of Peradeniya	2022 – 2024
Maths & Science Tutor , Arunella Education Welfare Programme	Jul – Dec 2023

References

Dr. Suneth Namal Karunarathna Senior Lecturer Department of Computer Engineering University of Peradeniya, Sri Lanka namal@eng.pdn.ac.lk	Mr. M.D. Rovin Shanila Associate Technical Lead WSO2 Pvt Ltd, Colombo 04 Sri Lanka rovinshanila@gmail.com
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